SOFTWARE ENGINEERING · TOPIC 1.2.4

Software Testing Definitions

“What is software testing?” — and why one small defect can cost a fortune.

Software Testing Module 1

Topic's Agenda

Unpacking the University Question: “What is software testing?”

01 Defining software testing — an art and a science

02 Why it matters more than ever in a digitized world

03 A real rocket that failed because of one small bug

04 The software testing process — as a flow and as a cycle

UQ: What is software testing?

MU — May 17, May 19

The art of investigation

Software testing is the art of investigating software to ensure its quality is in line with the requirements of the client.

The systematic science

It is carried out in a systematic manner with the intent of finding defects, and is required for evaluating the system.

Notice the word “art” and the word “systematic” in the same definition — testing needs both creative thinking and disciplined process.

Why This Matters More Than Ever

As technology advances, everything around you is getting digitized.

Banking from your phone

You can access your bank account online, anytime, from anywhere.

Shopping from home

You can shop from the comfort of your home — the options are endless.

What if it's defective?

Have you ever wondered what would happen if these systems turned out to be defective?

One Small Defect, One Huge Loss

“One small defect can cause a lot of financial loss. It is for this reason that software testing is now emerging as a very powerful field in IT.”

Ariane 5, Flight 501 · June 4, 1996

A single software module, reused without re-testing from the earlier Ariane 4 rocket, could not handle Ariane 5's faster flight data.

37	$370M	10
seconds after launch before the rocket self-destructed	value of the rocket and its satellite payload, lost	years of development work destroyed in an instant

*The lesson: the module worked perfectly for years on Ariane 4. Nobody re-tested it against Ariane 5's new flight parameters — an assumption, not a test, caused the **failure**.*

Software Doesn't Rust — But It Can Still Fail

No physical wear or tear

Unlike a physical product, software never rusts, corrodes, or mechanically wears out over time.

But design errors persist

Design errors written into the code can absolutely make your life difficult — if they go undetected.

Unlike a bridge that visibly rusts, a software defect gives no warning signs — it hides silently until the exact condition that triggers it finally occurs.

Ship With Bugs, Ship With Risk

Regular testing keeps software aligned with what the client actually asked for.

Finding the needle in a haystack

Scanning hundreds and thousands of lines of code to find and fix one bug is not an easy task — after release, it only gets harder.

Fixing one bug may create another

You never know that while fixing one bug, you may unknowingly introduce another bug elsewhere in the system.

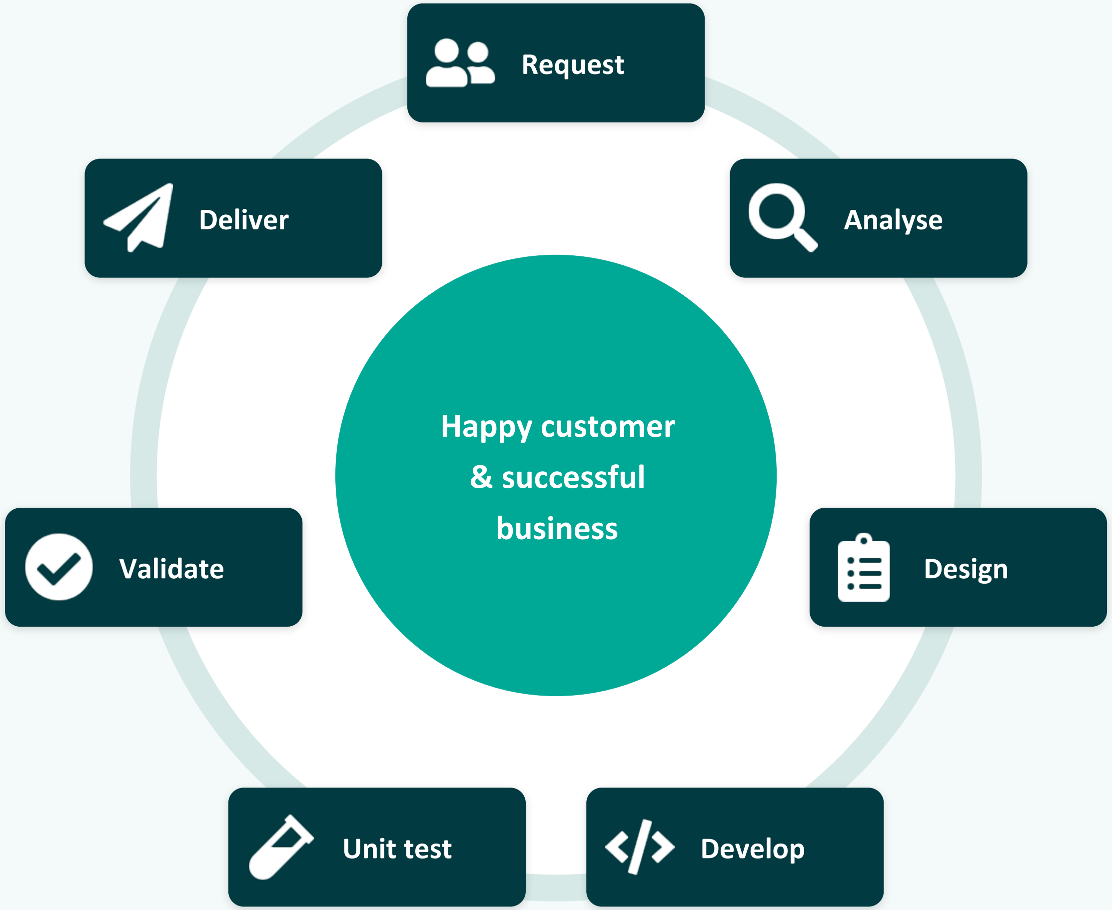
The Software Testing Process — A Linear View

Fig. 1.2.5 (part 1) — from customer request to delivery.

Customer requests	*Understand the requirements which will impart business value.*
Analyse	*Identify the features needed to fulfil the customer's requirements.*
Implement	*Design and code the solution to produce the required features.*
Integrate	*Unit test and integrate the solution to enable built-in quality.*
Test	*Validate both functional and non-functional aspects of the solution.*
Deliver	*Deliver the software and reap the benefits of success in business.*

The Software Testing Process — A Continuous Cycle

delivery isn't the end; it feeds the next request.



Notice: "Deliver" flows right back into a new "Request" — testing is a loop, not a finish line.

Key Takeaways

- Software testing is both an art of investigation and a systematic, defect-focused process.
- As everything digitizes, undetected defects can affect banking, shopping, and daily life.
- Ariane 5 lost $370M in 37 seconds because reused code was never re-tested for new conditions.
- Software never wears out physically, but undetected design errors can fail without warning.
- Testing is a continuous loop — delivery feeds the next request, aimed at a happy customer.